

3.3 Quadratic Programming

A convex quadratic with linear constraints. SVM is the running example.

These notes walk through the lecture slides. The original deck is unchanged; use **(slides)** on the course hub if you want that PDF.

The notes follow Deisenroth, Faisal, and Ong, *Mathematics for Machine Learning*.

1. Quadratic cost, linear constraints

A **quadratic program** keeps linear inequalities and lets the objective be quadratic. That is the next step past an LP: nonlinear models whose cost is a bowl, plus the same polyhedron as before.

This is also how **regularization** enters: a quadratic penalty on θ is still a QP when the constraints stay linear. Soft-margin **SVM** is the example this course cares about.

$$\min_{\theta \in \mathbb{R}^d} \frac{1}{2} \theta^\top Q \theta + \mathbf{c}^\top \theta \quad \text{subject to} \quad A\theta \leq \mathbf{b},$$

with $A \in \mathbb{R}^{m \times d}$, $\mathbf{b} \in \mathbb{R}^m$, $\mathbf{c} \in \mathbb{R}^d$, and $Q \in \mathbb{R}^{d \times d}$ symmetric **positive definite**. Then the objective is **convex**, so a local min is the unique global min. Still d variables and m linear constraints.

2. Dual

Lagrangian

$$\mathcal{L}(\theta; \lambda) = \frac{1}{2} \theta^\top Q \theta + \mathbf{c}^\top \theta + \lambda^\top (A\theta - \mathbf{b}).$$

Stationarity in θ :

$$Q\theta + (\mathbf{c} + A^\top \lambda) = 0.$$

If Q is invertible,

$$\theta = -Q^{-1}(\mathbf{c} + A^\top \lambda).$$

Substitute back. The dual Lagrangian is the concave quadratic

$$D(\lambda) = -\frac{1}{2}(\mathbf{c} + A^\top \lambda)^\top Q^{-1}(\mathbf{c} + A^\top \lambda) - \lambda^\top \mathbf{b}.$$

The dual problem is therefore

$$\max_{\lambda \in \mathbb{R}^m} -\frac{1}{2}(\mathbf{c} + A^\top \lambda)^\top Q^{-1}(\mathbf{c} + A^\top \lambda) - \lambda^\top \mathbf{b} \quad \text{subject to} \quad \lambda \geq 0.$$

Week 5 does the same substitution for the SVM dual, then replaces inner products by a kernel.

3. Why SVM is a QP

Soft-margin SVM:

$$\min_{\theta \in \mathbb{R}^l} \quad \frac{1}{2} \|\theta\|^2 + C \sum_{n=1}^N \xi_n \quad \text{subject to} \quad y_n(x_n^\top \theta + b) \geq 1 - \xi_n, \quad \xi_n \geq 0.$$

- θ : hyperplane weights.
- C : trade-off between margin size and slack.
- ξ_n : slack; a point may sit on the wrong side of the margin, at a linear cost.

Quadratic objective, linear inequalities: a **QP**, not an LP, because of $\|\theta\|^2$.

4. Practice

1. Why is a soft-margin SVM a quadratic program?
2. What does Q positive definite buy you that a general quadratic objective does not?
3. After $\theta = -Q^{-1}(c + A^\top \lambda)$, what variables does the dual still optimize, and what constraint remains on them?